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**DESIGN OF A LOW-COST REAL-TIME INTELLIGENT FUEL
ASSET MANAGEMENT SYSTEM**



Background Info

Market Infrastructure

4,270

Active fuel stations nationwide

2,762

Retail stations classified in tiers

Fuel Consumption

5.84 million metric tons annual consumption (2024/2025)

46% diesel, 30% petrol distribution

Monitoring Reliability Gap

±10mm Inherent Error

95% Detection Probability

90% Technology Vacuum

Economic leakage statistics: The Hidden Costs

Theft & Siphoning

3 - 5%
of operational
revenue lost at
independent stations

1



Ghost Deliveries

1.64 - 4.7%
short-filling during
tanker offloading—
millions lost to
undetected under-
delivery

2



Thermal Variance

1% per 9.4°C
without 15.5°C
standardization, owners
"buy volume" but "sell
energy," losing on cooler
mornings

3

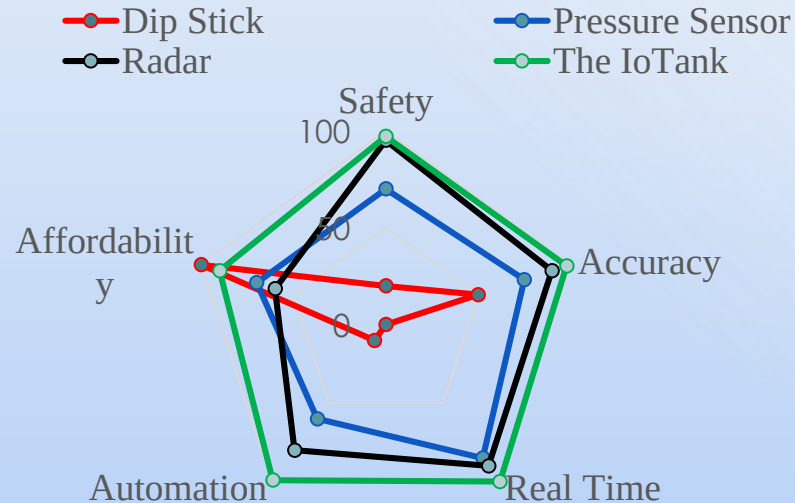


The Problem & Opportunity

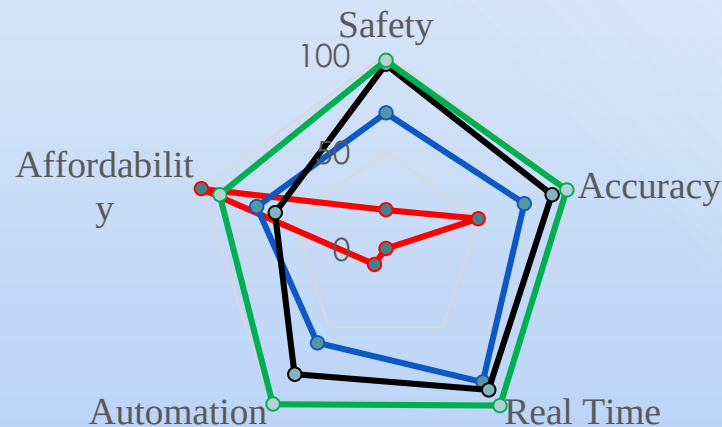
Fuel Level Monitoring Methods

1 Manual Dipstick / Dip Tape

Highly prone to human error
Inaccuracy
Insufficient leak-detection standards.



—●— Dip Stick
—●— Radar
—●— Pressure Sensor
—●— The IoTank



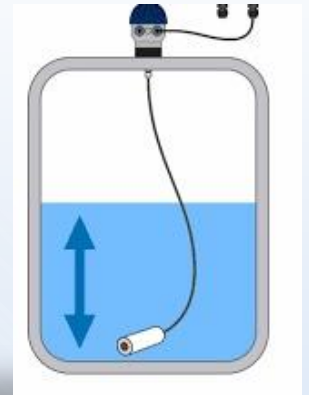
AIM

Lower cost on accuracy and automation of a safe Fuel Monitoring and Inventory system

2

Hydrostatic & Pressure Sensors

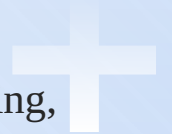
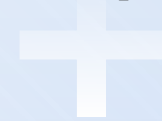
Accuracy affected by fuel density changes (temperature shifts).



Radar Gauges

3

Costly
Volume discrepancy



The Core Problem

- Inaccurate inventory management and planning.
- Costly systems.
- Small, Hard-to-Detect Leaks undetected
- Volume Discrepancy for uneven Tanks & Gas tank tilting, thermal expansion
- Safety risks.
- Failed Theft detection

The IoTank Solution Overview

System Vision: Hybrid Edge-to-Cloud Architecture

The system is split into specialized layers to
★maximize speed ★security ★intelligence

Key Differentiator: Volume-First, AI-Powered

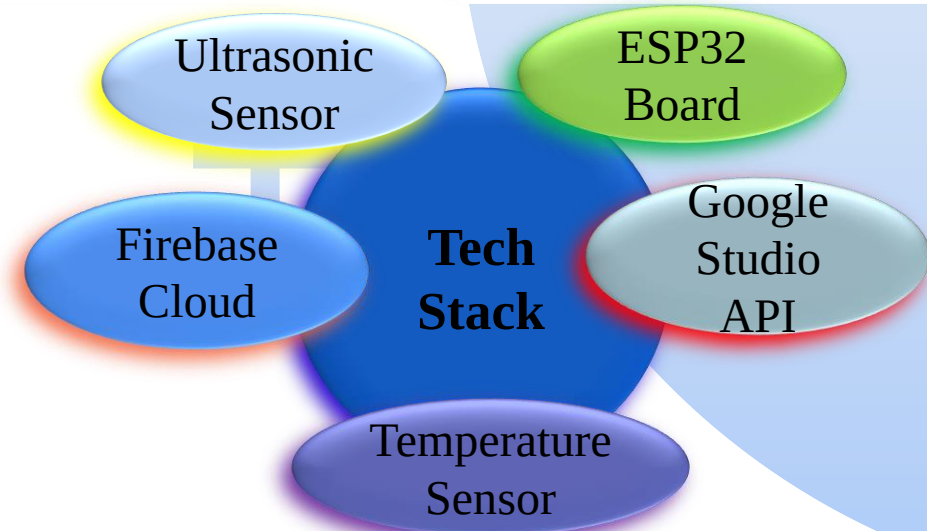
We focus on delivering real-time Liters-not confusing distance readings- allowing the AI layer to immediately perform accurate high-value financial and predictive tasks that translates sensor data directly into actionable business intelligence.

01 Edge (Fuel Station)

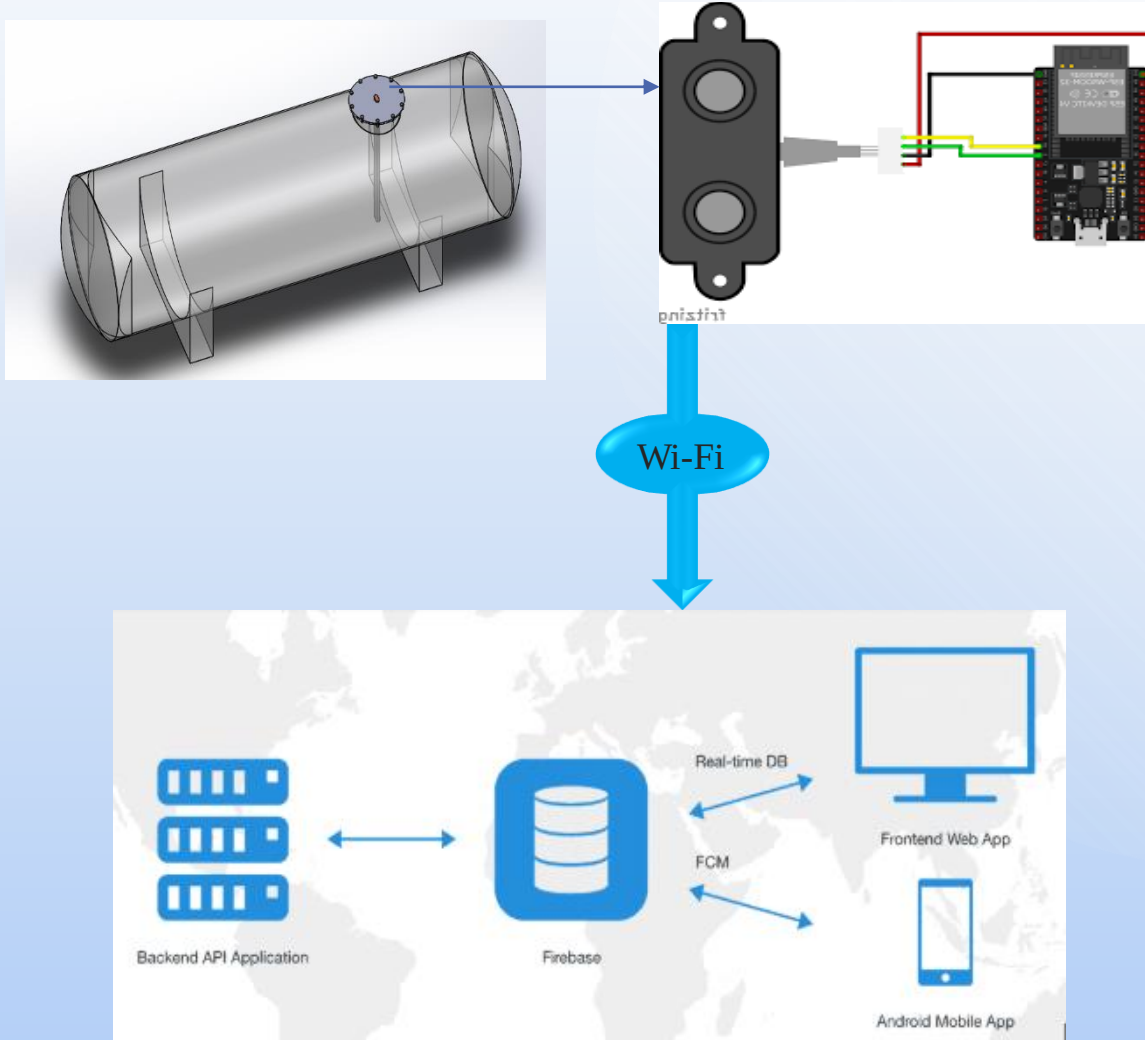
- ❖ High-speed
- ❖ low-latency data acquisition
- ❖ volume calculation.

02 Cloud (Firebase)

- ❖ Cloud dashboards for remote monitoring.
- ❖ Secure data storage
- ❖ Complex AI analysis
- ❖ User interface hosting.



Core Components



1

Edge

ESP32 Microcontroller +
A02YYUW Ultrasonic Sensor.

2

Backend

Google Cloud Functions + Firebase
Firestore (Zero-Cost, Scalable).

3

Intelligence

Google Studio API (Gemini).

4

Frontend

Web Dashboard- Real-time and
Predictive Analytics.

Safety & Edge Reliability Design

Hardware & Safety Isolation Layer

1

Zone 0 Isolation

- Acoustically transparent chemical barrier using <0.13mm PTFE membrane
- Complete vapor separation from electronic components
- Chemically inert in petroleum environments

2

Static Dissipation

- Conductive PVDF (Kynar) guide tube drains electrostatic charges
- Prevents spark generation in explosive atmospheres
- Grounded pathway for charge dissipation

3

Environmental Control

- Guide tube (stilling well) eliminates surface turbulence
- Directs focused 40kHz ultrasonic waves
- Minimizes acoustic interference and signal scattering

Safety Instrumented System (SIS)

4

Electrical Isolation

- Opto-isolator circuit for UART communication lines
- Zero electrical path between tank and ESP32
- Galvanic separation prevents current flow to hazardous zone

5

Energy Limiting

- Series current limiter restricts maximum energy delivery
- Fast-blow fuse as secondary protection layer
- Total energy maintained below 0.2 mJ (gasoline MIE threshold)

6

Critical Kill-Switch

- P-Channel MOSFET physically cuts transducer power
- DS18B20 temperature sensor triggers at 60°C threshold
- Fail-safe design prevents operation in unsafe thermal conditions



7

Core Correction Equations

Because the ESP32 receives a pre-calculated distance from the A02YYUW, we must reverse-engineer the Time of Flight to account for changing Gas constant (R) for accurate fuel measurements.

**True Acoustic Velocity
(v true)**

$$v_{true} = \sqrt{\frac{\gamma \cdot R \cdot T}{M}}$$

Undoing the sensor's hardcoded 348 m/s error whereby ultrasonic wave speeds are mostly varied by Temperatures, vapor density and

$$D_{compensated} = D_{raw} \times \left(\frac{v_{true}}{348} \right)$$

Dynamic Molar Mass



Using the Newton-Laplace formula to account for temperature (T in Kelvin) and the changing molar mass (M) of fuel vapours:

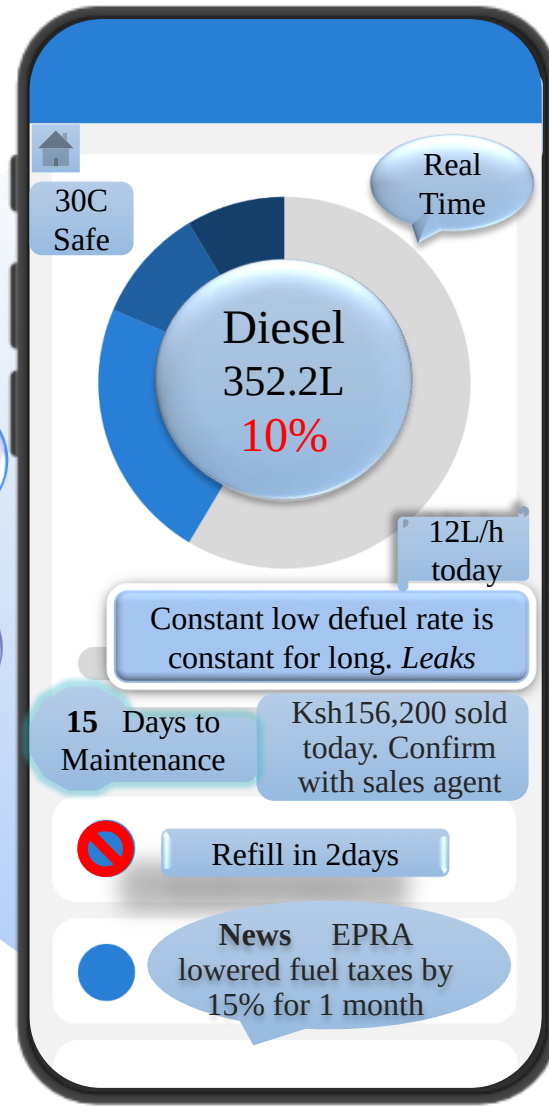
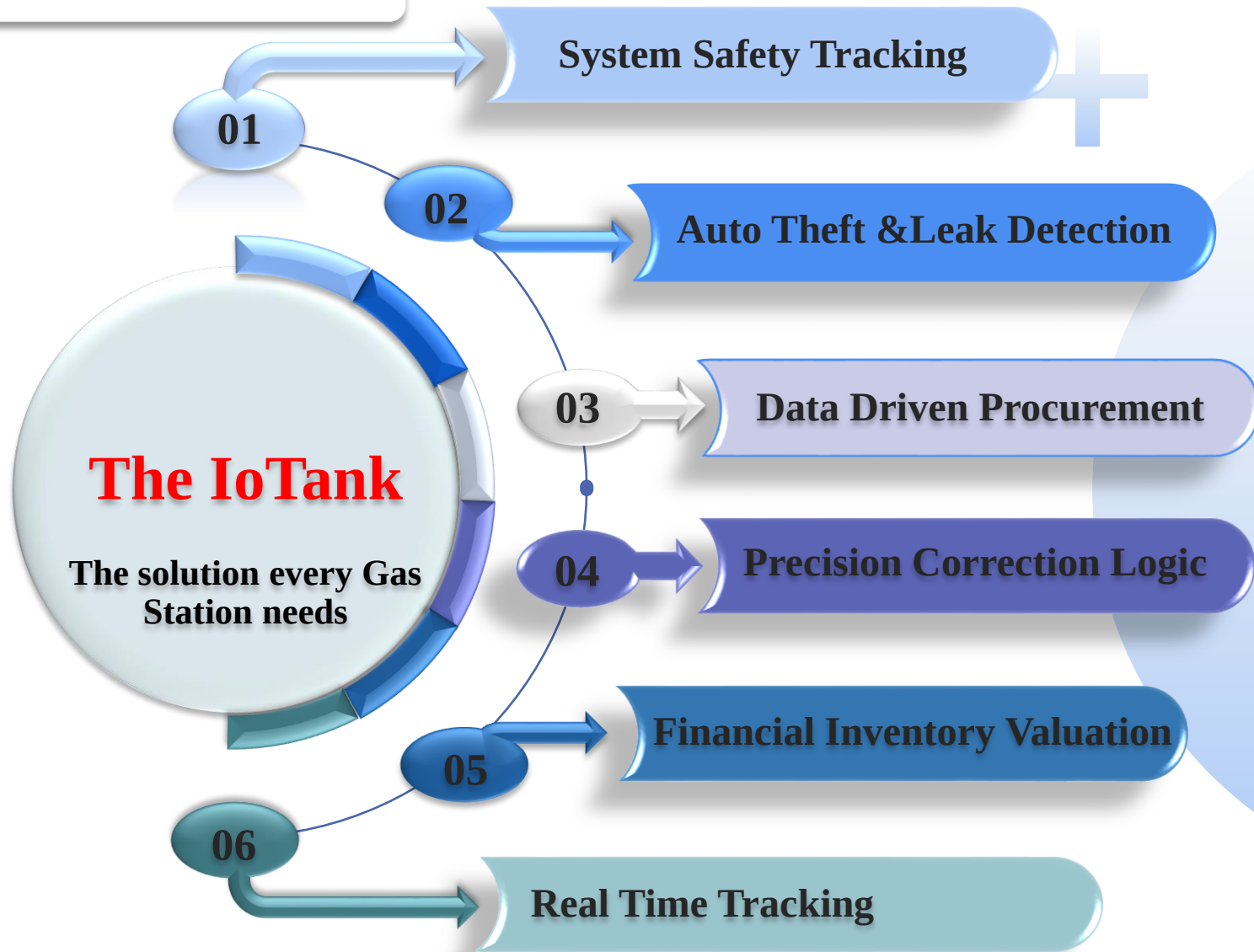


**Compensated Distance
(D compensated):**



Implemented via a Lookup Table on the ESP32, adjusting M from 0.029 kg/mol (High Level) to ≈ 0.105 kg/mol (Low Level/Saturated Vapor).

The IoTank Hub



The IoTank web



Core Focus Implementation Summary

Safety

- ✓ PTFE/PVDF Isolation
- ✓ Zone 0 Compliance
- ✓ Thermal Kill-Switch
- ✓ Intrinsically Safe Design

Low Cost

- ✓ Total budget: ksh 20,000
- ✓ Off-the-shelf IoT components
- ✓ Cloud-based infrastructure
- ✓ Accessible to small operators

Accuracy

- ✓ Newton-Laplace vapor
- ✓ Correction
- ✓ Temperature standardization (15.5°C)
- ✓ $\pm 1\text{mm}$ precision measurement
- ✓ $<0.2\%$ variance vs 7% traditional

Automation

- ✓ 24/7 continuous monitoring
- ✓ AI-driven procurement advice
- ✓ Automated theft/leak alerts
- ✓ Self-calibrating algorithms

Real Time

- ✓ Sub-500ms edge-to-cloud latency
- ✓ Live 3D Digital Twin visualization
- ✓ Instant theft detection
- ✓ Live market-based valuation



Budget

Quarter	Descriptions	Units	Price(Ksh)
Arduino	ESP32	1	2,500
Ultrasonic Sensor	A02YYUW	1	3,000
Guide Tube and floater	HDPE + Teflon	1	1,500
Fuel Tank	Plastic Model	1	1,000
Arduino cover	Water proof & Isolation Barrier	1	2,000
Web Dev & Cloud	Coding	30 days	3,000
Power Supply	5V	1	1,000
Miscellaneous		0 - infinity	4,000
Total			ksh 18,000

References

To support the technical and legal validity of your IoTank project, here are seven high-quality references. These cover the physics of sound, industrial safety standards for fuel, and the regulatory requirements for tank monitoring.

- ❖ ASTM International. (2008). *ASTM D1250-08: Standard Guide for the Use of the Joint API and ASTM Petroleum Measurement Tables*.
- ❖ National Fire Protection Association. (2021). *NFPA 30: Flammable and Combustible Liquids Code*. *
- ❖ U.S. Environmental Protection Agency (EPA). (2015). *40 CFR Part 280: Technical Standards and Corrective Action Requirements for Owners and Operators of Underground Storage Tanks (UST)*.
- ❖ Kinsler, L. E., Frey, A. R., Coppens, A. B., & Sanders, J. P. (2000). *Fundamentals of Acoustics* (4th ed.). Wiley.
- ❖ Espressif Systems. (2023). *ESP32-WROOM-32 Series Datasheet*.
- ❖ American Petroleum Institute (API). (2017). *API Manual of Petroleum Measurement Standards (MPMS) Chapter 3: Tank Gauging*.
- ❖ International Electrotechnical Commission. (2017). *IEC 60079-11: Explosive Atmospheres - Part 11: Equipment Protection by Intrinsic Safety "i"*.